STPM Physics · Semester 3 · Chapter 24

Quantum Physics

24.1 Photon

Definitions & laws

- **Photon** is a quantum (packet) of electromagnetic radiation of energy E = hf.
- **Photoelectric effect** is the emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency falls on it.
- **Work function φ** is the minimum energy required to remove an electron from the surface of a metal; **threshold frequency f_0** is the minimum frequency for emission, $f_0 = \frac{\varphi}{h}$.
- **Stopping potential V_s** is the minimum reverse potential difference that reduces the photocurrent to zero; $eV_s = K_{max}$.
- **Einstein's photoelectric equation**: $hf = \varphi + K_{max}$, an electron absorbs one photon whose energy is shared between the work function and the kinetic energy.
- Observations that wave theory cannot explain: a threshold frequency exists; K_{max} is independent of intensity and depends on frequency; emission is instantaneous.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Photon energy	$E = hf = \frac{hc}{\lambda}$	E = photon energy (J); f = frequency (Hz); λ = wavelength (m); h = 6.63 × 10^{-34} J s; c = 3.00 × 10^{8} m s^{-1}
Photoelectric equation	$hf = \varphi + ½mv^2_max$; $\varphi = hf_0$	f = frequency of incident radiation (Hz); φ = work function (J); f_0 = threshold frequency (Hz); m = electron mass (kg); v_{max} = maximum speed of photoelectrons (m s^{-1})
Stopping potential	$eV_s = ½mv^2_max = hf - \varphi$	V_s = stopping potential (V); e = electron charge (C); m = electron mass (kg); v_{max} = maximum speed of photoelectrons (m s^{-1}); φ = work function (J)
Photon momentum	$p = \frac{h}{\lambda}$	p = photon momentum (kg m s^{-1}); h = Planck constant (J s); λ = wavelength (m)
Electronvolt	1 eV = 1.60 × 10^{-19} J; $E\ (eV) \approx \frac{1240}{\lambda}$ (nm)	E (eV) = photon energy in electronvolts; λ (nm) = wavelength in nanometres

- Graph of K_{max} against f: straight line, gradient = h, intercept on the f-axis = f_0, intercept on the energy axis = −φ.

How to answer (MPM style)

- Write E = hf or $\frac{hc}{\lambda}$, convert nm to m and eV to J (or J to eV) consistently, then substitute with units.
- For photoelectric questions write hf = φ + K_{max}, then K_{max} = hf − φ; convert both energies to the same unit before subtracting.
- For explanations: state that intensity ∝ number of photons per second ∝ number of photoelectrons per second, and K_{max} depends on frequency only.
- Say 'below the threshold frequency no emission occurs, no matter how intense the light'.

Section A traps

Trap: Increasing intensity increases the number of photoelectrons (the saturation current), not K_{max} and not the stopping potential.

Trap: Subtracting energies in different units: hf = 4.64 × 10^{-19} J = 2.90 eV; φ = 2.3 eV gives K_{max} = 0.60 eV. Giving 2.90 eV (forgetting φ) or 5.20 eV (adding φ) are the usual wrong options.

Trap: Threshold wavelength $\lambda_0 = \frac{hc}{\varphi}$ is the LONGEST wavelength that causes emission; shorter wavelengths emit electrons. Do not reverse the inequality.

Trap: Reporting hf (in joules) when the options are in electron-volts, or stating the stopping potential in eV (it is in volts: $V_s = \frac{K_{max}}{e}$).

Tricks

Trick: Use E (eV) ≈ $\frac{1240}{\lambda}$ (nm) for fast checks: 632.8 nm gives about 1.96 eV; 450 nm gives about 2.76 eV (4.42 × 10^{-19} J).

Trick: When K_{max} is in eV, the stopping potential in volts has the same numerical value.

24.2 Wave-Particle Duality

Definitions & laws

- **Wave-particle duality**: radiation and matter exhibit both wave properties (diffraction, interference) and particle properties (photoelectric effect, photon momentum) depending on the experiment.
- **de Broglie hypothesis**: a moving particle of momentum p has an associated wavelength $\lambda = \frac{h}{p}$.
- Wave nature of electrons is shown by electron diffraction through a thin crystal or graphite film (Davisson–Germer experiment): a ring (diffraction) pattern is formed.
- Particle nature of light is shown by the photoelectric effect; wave nature of light by diffraction and interference.

- The wave behaviour of macroscopic objects is not observed because the de Broglie wavelength is far too small compared with any obstacle or gap.

Formulae

Quantity / law	Formula	Symbols, units & conditions
de Broglie wavelength	$\lambda = \frac{h}{p} = \frac{h}{mv}$	λ = de Broglie wavelength (m); h = Planck constant (J s); p = momentum (kg m s^{-1}); m = mass (kg); v = speed (m s^{-1})
Wavelength from kinetic energy	$p = \sqrt{2mK}$; $\lambda = \frac{h}{\sqrt{2mK}}$	K = kinetic energy (J); m = mass (kg); p = momentum (kg m s^{-1}); λ = wavelength (m)
Electron accelerated through a potential difference	$eV = \frac{1}{2}mv^2$; $\lambda = \frac{h}{\sqrt{2meV}}$	V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed (m s^{-1}); λ = wavelength (m)

- Constants: h = 6.63×10^{-34} J s, $m_e = 9.11 \times 10^{-31}$ kg, e = 1.60×10^{-19} C, $m_p = 1.67 \times 10^{-27}$ kg.

How to answer (MPM style)

- Write $\lambda = \frac{h}{mv}$ (or $\frac{h}{\sqrt{2meV}}$), substitute with SI units and give the answer in metres to 3 s.f. (2 s.f. if the data are 2 s.f.).
- For evidence of both natures name the experiment AND the property shown: diffraction/interference → wave; photoelectric effect → particle.
- For an electron accelerated through V, first find the speed or momentum from $eV = \frac{1}{2}mv^2$, then $\lambda = \frac{h}{p}$.
- State: a larger momentum gives a shorter wavelength ($\lambda \propto \frac{1}{p}$).

Section A traps

Trap: Saying that the photoelectric effect alone shows both natures: it demonstrates the particle nature only. Two different phenomena are needed.

Trap: Using the speed of light for the electron or confusing the mass of a proton (1.67×10^{-27} kg) with that of an electron. A proton at the same speed has a wavelength about 1800 times shorter.

Trap: Forgetting the square root: $\lambda = \frac{h}{\sqrt{2meV}}$, not $\frac{h}{2meV}$. For V = 150 V, $\lambda \approx 1.0 \times 10^{-10}$ m.

Trap: Rutherford scattering and the Millikan experiment do not show the wave nature of the electron.

Tricks

Trick: Fast electron wavelength: λ (nm) ≈ $\frac{1.23}{\sqrt{V}}$ (V in volts). For 150 V, λ ≈ 0.100 nm.

> **Trick:** Same speed, heavier particle: $\lambda \propto \frac{1}{m}$, so a proton's wavelength is $\frac{1}{1836}$ that of an electron.

24.3 Atomic Structure

Definitions & laws

- **Line spectrum** consists of discrete bright lines (emission) or dark lines (absorption) of definite wavelengths, characteristic of the element; it shows that electrons in atoms occupy discrete energy levels.
- **Bohr's postulates**: electrons move in certain stable orbits without radiating energy (stationary states); the angular momentum of an orbiting electron is quantised, $mvr = \frac{nh}{2\pi}$; radiation is emitted or absorbed only when an electron makes a transition between two levels, $hf = E_i - E_f$.
- **Ionisation energy** is the minimum energy needed to remove an electron from the ground state of an atom; for hydrogen it is 13.6 eV.
- **Spectral series of hydrogen**: Lyman (transitions to n = 1, ultraviolet), Balmer (to n = 2, visible), Paschen (to n = 3, infrared).
- **Laser action**: stimulated emission produces photons identical in phase, frequency and direction to the stimulating photon; population inversion is needed for light amplification.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Hydrogen energy levels	$E_n = -\frac{13.6}{n^2}$ eV	n = 1, 2, 3, … (n = 1 is the ground state); E_n = energy of level n (eV); 1 eV = 1.60×10^{-19} J
Photon emitted or absorbed in a transition	$hf = \frac{hc}{\lambda} = E_i - E_f$	E_i = energy of the initial (higher) level (J); E_f = energy of the final level (J); f = photon frequency (Hz); λ = wavelength (m)
Rydberg formula	$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$	R = 1.10×10^7 m^{-1} (Rydberg constant); λ = wavelength (m); n_i = initial level; n_f = final level
Bohr radius of orbit	$r_n = n^2 a_0$	r_n = radius of the nth orbit (m); a_0 = 5.29×10^{-11} m (Bohr radius); n = 1, 2, 3, …

How to answer (MPM style)

- Write the energies of both levels using $-\frac{13.6}{n^2}$, subtract (higher − lower) to obtain a positive photon energy, convert eV to J, then $\lambda = \frac{hc}{E}$.
- Explain line spectra in 2 points: electrons have only discrete energy levels; photons emitted in transitions have energy equal to the difference of two levels, so only specific wavelengths appear.
- Name the series (Lyman/Balmer/Paschen) and the region of the spectrum when asked.

- Give the wavelength in nm to 3 s.f. and the energy in eV.

Section A traps

Trap: Energy levels are negative; n = 2 (−3.40 eV) is higher than n = 1 (−13.6 eV). The photon energy for n = 2 → 1 is 10.2 eV, not 3.40 eV (which is the energy of the n = 2 level itself) or 13.6 eV (the ionisation energy).

Trap: Mistaking the Balmer wavelengths: n = 3 → 2 is 658 nm (red) and n = 5 → 2 is 435 nm (violet); 122 nm is the n = 2 → 1 (Lyman) line, in the ultraviolet.

Trap: Writing 'electrons move in continuous orbits so any wavelength is emitted'; the quantisation of energy levels is the point.

Trap: Dividing the photon energy in eV by hc directly without converting to joules.

Tricks

Trick: Shorter wavelength = larger energy gap. Lyman lines are shortest, then Balmer, then Paschen.

Trick: Minimum photon wavelength to ionise ground-state hydrogen: $\lambda = \frac{hc}{13.6}$ eV = 91 nm.

24.4 X-rays

Definitions & laws

- **X-rays** are electromagnetic waves of very short wavelength (about 10^{-11} to 10^{-9} m), produced when fast electrons are suddenly stopped by a metal target.
- **Continuous spectrum** arises from the deceleration of electrons in the target (bremsstrahlung); it has a sharp cut-off at the minimum wavelength λ_{min}, corresponding to the whole kinetic energy of an electron being converted to a single photon.
- **Characteristic spectrum** consists of sharp lines produced when an incident electron ejects an inner-shell electron of the target atom and an outer electron falls to fill the vacancy; the lines are characteristic of the target material.
- **Bragg's law**: constructive interference occurs for X-rays reflected from successive crystal planes when the path difference 2d sin θ is a whole number of wavelengths.
- Intensity of the X-ray beam depends on the tube current; the penetrating power (hardness) depends on the accelerating voltage.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Kinetic energy of electron at target	$eV = \frac{1}{2}mv^2$	electron reaching the target from rest; V = accelerating potential difference (V); e = electron charge (C); m = electron mass (kg); v = speed at the target (m s^{-1})
Minimum wavelength	$eV = \frac{hc}{\lambda_{min}}$; $\lambda_{min} = \frac{hc}{eV}$	λ_{min} = minimum wavelength (m); h = 6.63 × 10^{-34} J s; c = 3.00 × 10^{8} m s^{-1}; e = 1.60 × 10^{-19} C; V = accelerating potential difference (V)
Bragg's law	$2d \sin\theta = n\lambda$	d = spacing of the crystal planes (m); θ = glancing angle (measured from the planes, not the normal); n = 1, 2, 3, … (order); λ = wavelength (m)
Attenuation of X-rays	$I = I_0 e^{-\mu x}$	I_0 = incident intensity (W m^{-2}); I = transmitted intensity (W m^{-2}); μ = linear attenuation coefficient (m^{-1}); x = thickness of the absorber (m)

How to answer (MPM style)

- For λ_{min} write $\lambda_{min} = \frac{hc}{eV}$, substitute with units; state that λ_{min} depends only on V and not on the target material.
- For Bragg's law, state that θ is the glancing angle between the beam and the crystal planes; write $2d \sin\theta = n\lambda$ and solve.
- For the characteristic lines, state 'transition of an outer electron to a vacancy in an inner shell'.
- When asked for the electron speed, equate eV to $\frac{1}{2}mv^2$ and convert kV to V.

Section A traps

Trap: Using V in kV directly: 25 kV = 2.5 × 10^{4} V. Forgetting this gives λ_{min} a factor of 1000 error (options such as 3.1 × 10^{-10} m for 40 kV).

Trap: Doubling V halves λ_{min} ($\lambda_{min} \propto \frac{1}{V}$); options that double λ_{min} or leave it unchanged are wrong.

Trap: Using the angle from the normal in Bragg's law: θ is the glancing angle measured from the crystal planes.

Trap: Saying the continuous spectrum is characteristic of the target; only the sharp lines depend on the target material.

Tricks

Trick: Quick estimate: λ_{min} (nm) ≈ $\frac{1.24}{V}$ (kV). For 25 kV, $\frac{1.24}{25}$ = 0.0496 nm = 5.0 × 10^{-11} m.

> **Trick:** Bragg: $\sin\theta \le 1$, so the wavelength must satisfy $\lambda \le \frac{2d}{n}$ for order n.

24.5 Nanoscience

Definitions & laws

- **Nanoscience** is the study of materials and structures with at least one dimension of the order of 1 nm to 100 nm.
- **Quantum confinement**: when the size of a particle becomes comparable with the de Broglie wavelength of its electrons, the energy levels become discrete and their spacing increases as the size decreases.
- **Quantum dot** is a nanoscale semiconductor crystal in which electrons are confined in all three dimensions; smaller dots have a larger energy gap and emit shorter-wavelength (bluer) light.
- The surface-area-to-volume ratio of nanoparticles is very large, so a large proportion of atoms lie at the surface, which gives a high chemical reactivity and catalytic activity.
- **Scanning tunnelling microscope (STM)** uses the quantum-mechanical tunnelling current between a sharp conducting tip and the surface; the current depends very sensitively (exponentially) on the tip–surface separation, so atoms can be imaged.
- Examples: carbon nanotubes (high strength, electrical conduction), graphene, nanoparticles in sunscreens and catalysts.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Surface area to volume ratio of a sphere	$\frac{3}{r}$	r = radius of the sphere (m); ratio in m^{-1}; the ratio increases as the size decreases
Surface area to volume ratio of a cube	$\frac{6}{L}$	L = side of the cube (m); ratio in m^{-1}; the ratio increases as the size decreases
Particle in a one-dimensional box	$E_n = \frac{n^2h^2}{8mL^2}$; $E \propto \frac{1}{L^2}$	E_n = energy of level n (J); n = 1, 2, 3, ...; h = Planck constant (J s); m = mass of the particle (kg); L = width of the box (m)
Energy gap and wavelength	$E_{gap} = \frac{hc}{\lambda}$	E_{gap} = energy gap (J); h = Planck constant (J s); c = speed of light ($m\ s^{-1}$); λ = emitted wavelength (m); a larger gap corresponds to a shorter emitted wavelength

- 1 nm = 10^{-9} m; the nanoscale is about 1–100 nm.

How to answer (MPM style)

- For size effects state: smaller size → energy levels more widely spaced → larger bandgap → emission or absorption at a shorter wavelength.
- For reactivity state: smaller particles have a larger surface-area-to-volume ratio, so more atoms are exposed at the surface.
- For STM state: tunnelling current → very sensitive to the separation → used to map the surface atom by atom.
- Give the nanoscale range as 1 nm to 100 nm.

Section A traps

Trap: Saying the bandgap DEcreases as the particle gets smaller; it increases, so the colour shifts towards blue (shorter wavelength).

Trap: Quoting the nanoscale as 1 µm to 100 µm or 1 pm to 100 pm; the nanoscale is 1–100 nm.

Trap: Saying that nanoparticles obey different chemistry or have fewer atoms 'so each atom is more reactive'; the reason is the large surface-area-to-volume ratio.

Trap: Attributing quantum dot colour to the number of atoms alone: bigger dots (smaller gap) emit redder light, not bluer.

Tricks

Trick: For quantum dots: small → blue, large → red. The gap is inversely related to the wavelength emitted.

Trick: For surface-area-to-volume ratio: halving the radius doubles the ratio $\frac{3}{r}$.